Lab Report: Installing Debian Linux in Hyper-V

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Lab Objective

The primary objective of this lab was to successfully create and configure a nested virtual machine (VM) running Debian Linux within Hyper-V.

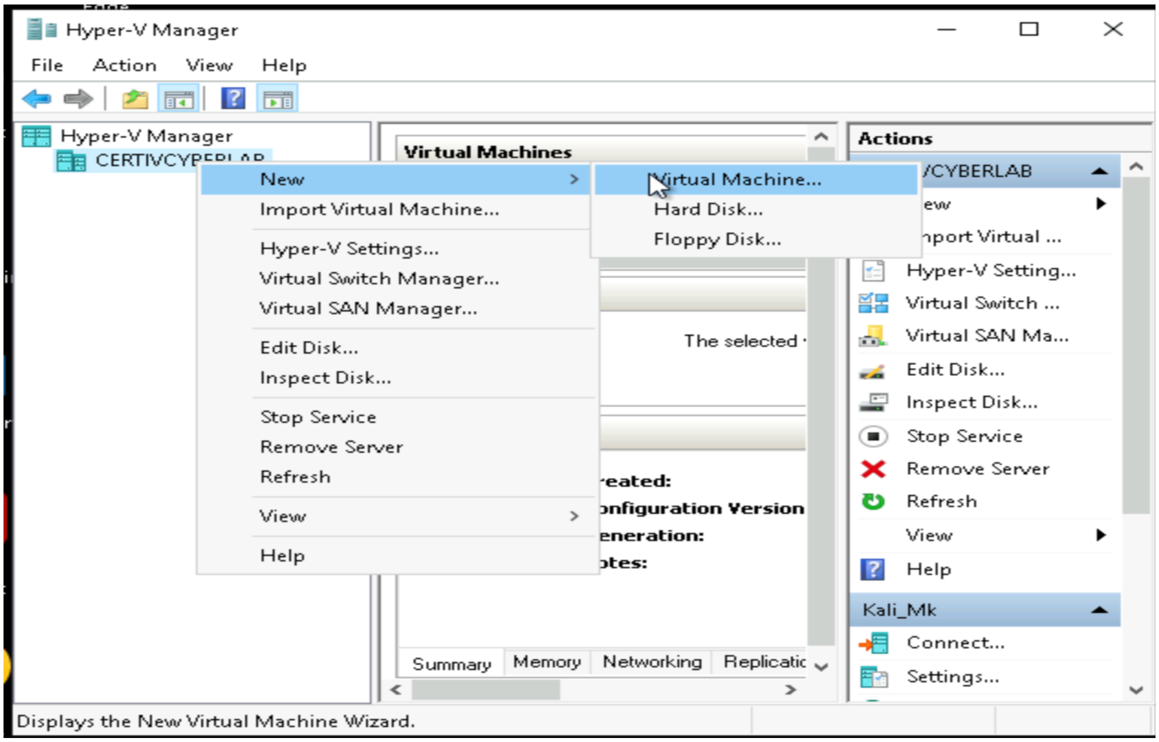
Tools / Environment Used
- Hyper-V Manager (Windows)
- Debian Linux VHDX (pre-existing virtual hard disk)
- Cert_IV_Cyber_Virtual_Switch (custom virtual switch)
- Firefox (within Debian Linux)

Step-by-Step Tasks Performed

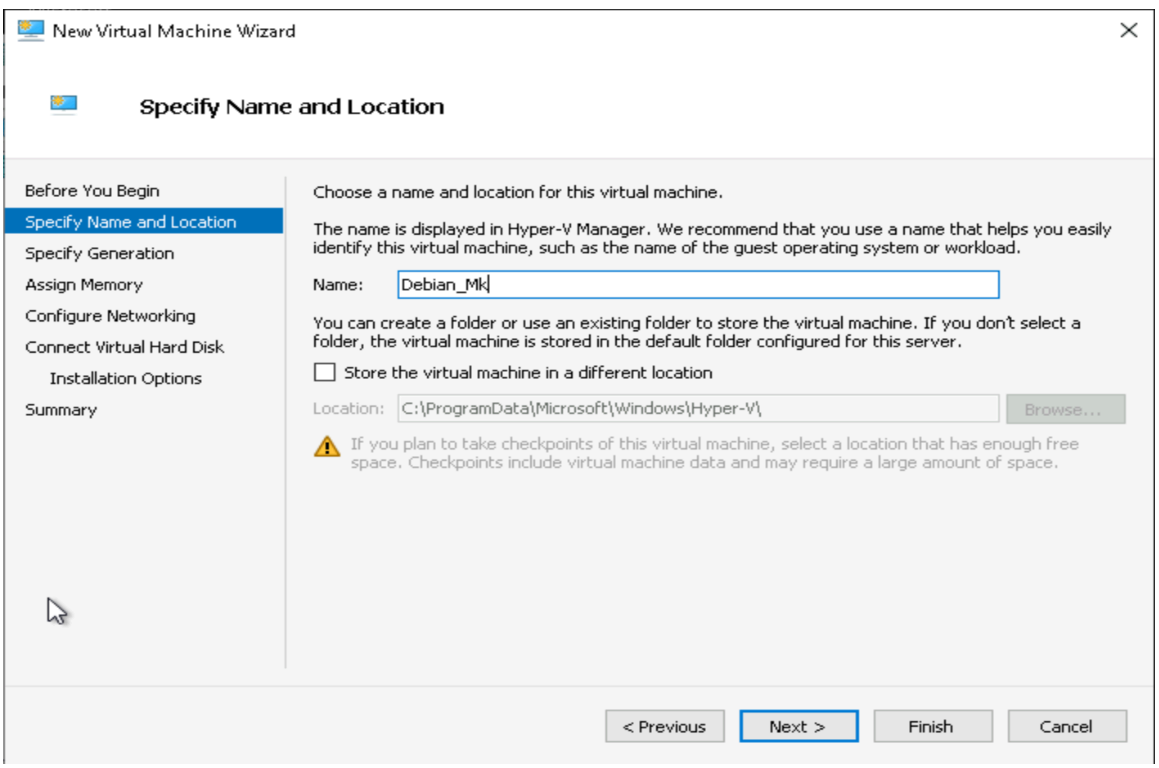
Task 1: Creating a Nested VM using Existing ISO

Step 1 – Creating the Virtual Machine

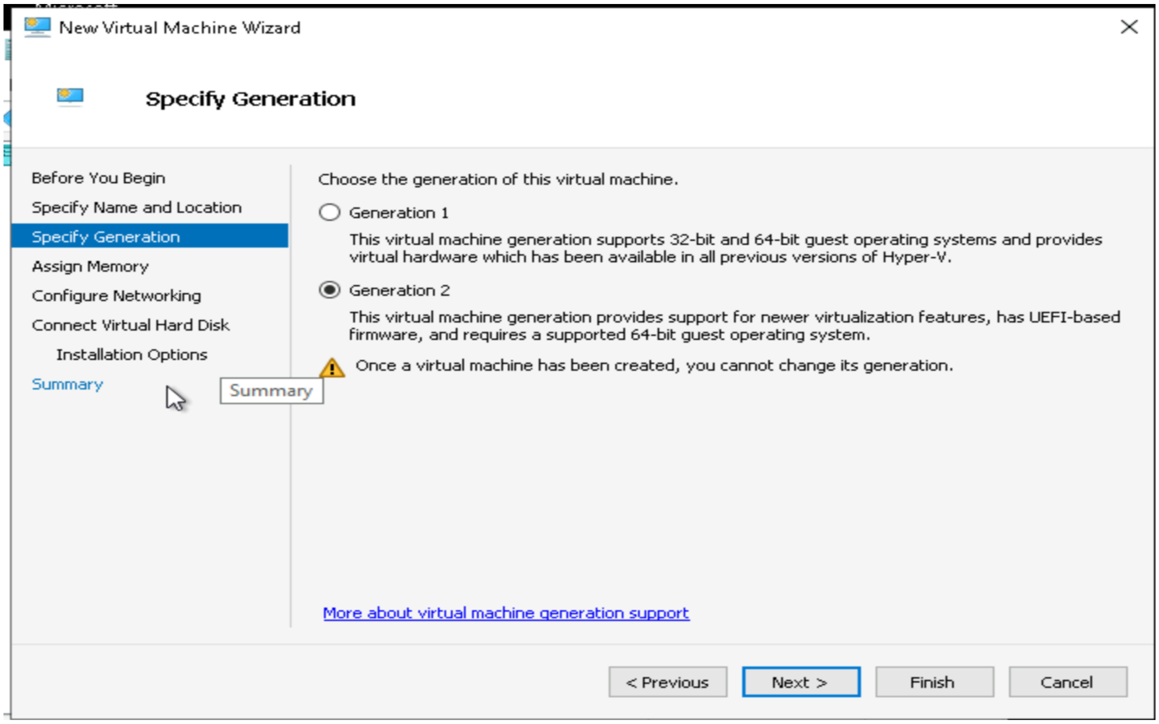
- Opened Hyper-V Manager.
- Created a new virtual machine.



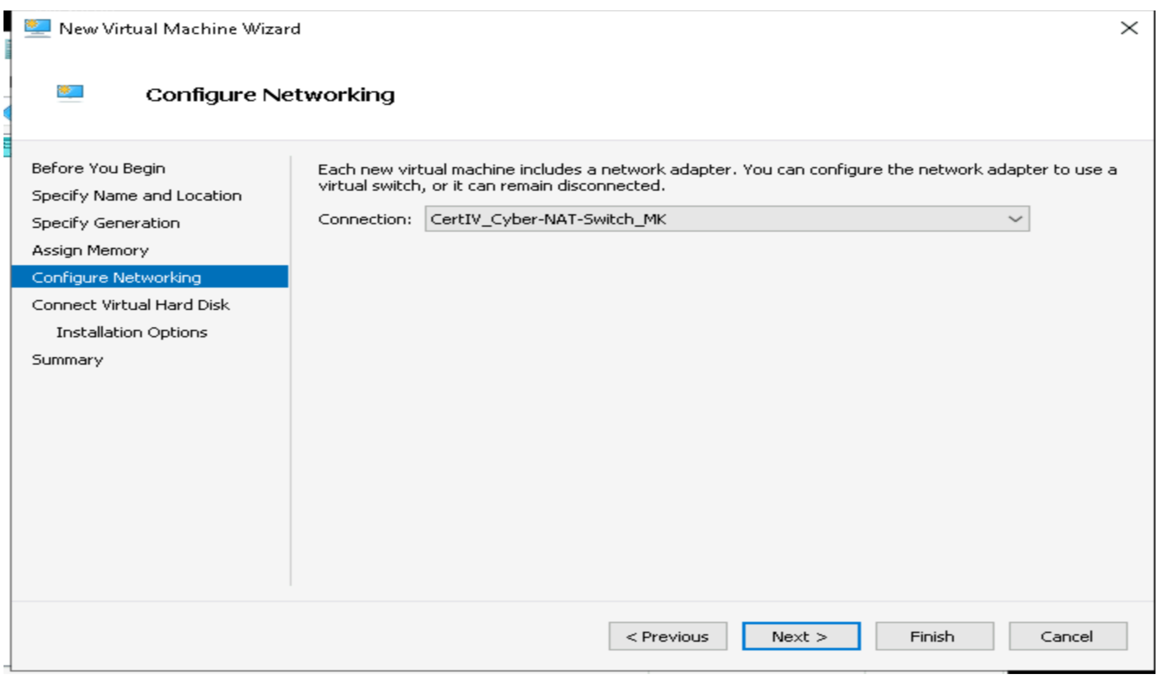
- Named the VM: Debian_[YourInitials].



- Selected Generation 2.



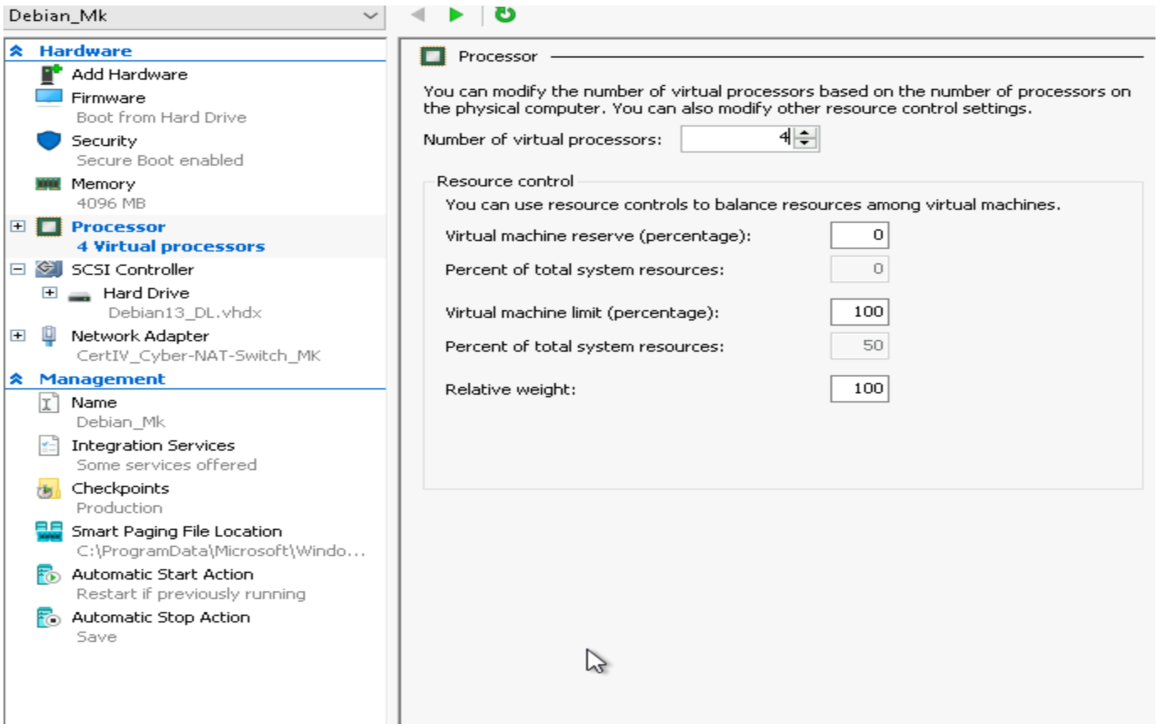
- Assigned 4096 MB RAM and disabled Dynamic Memory.
- Selected the correct virtual switch.



- Attached Debian.vhdx.
- Completed the wizard.

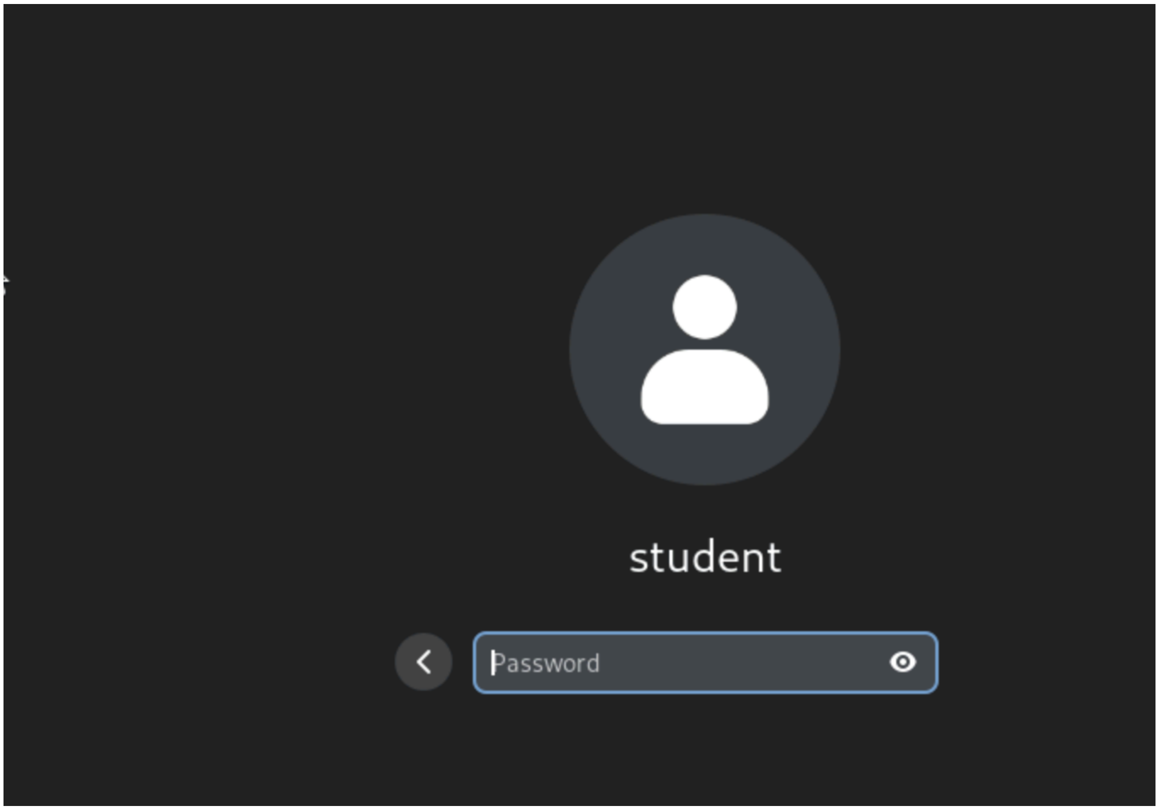
Step 2 – Adjusting VM Settings

- Increased virtual processors to 4.
- Disabled Secure Boot.
- Enabled Guest Services.



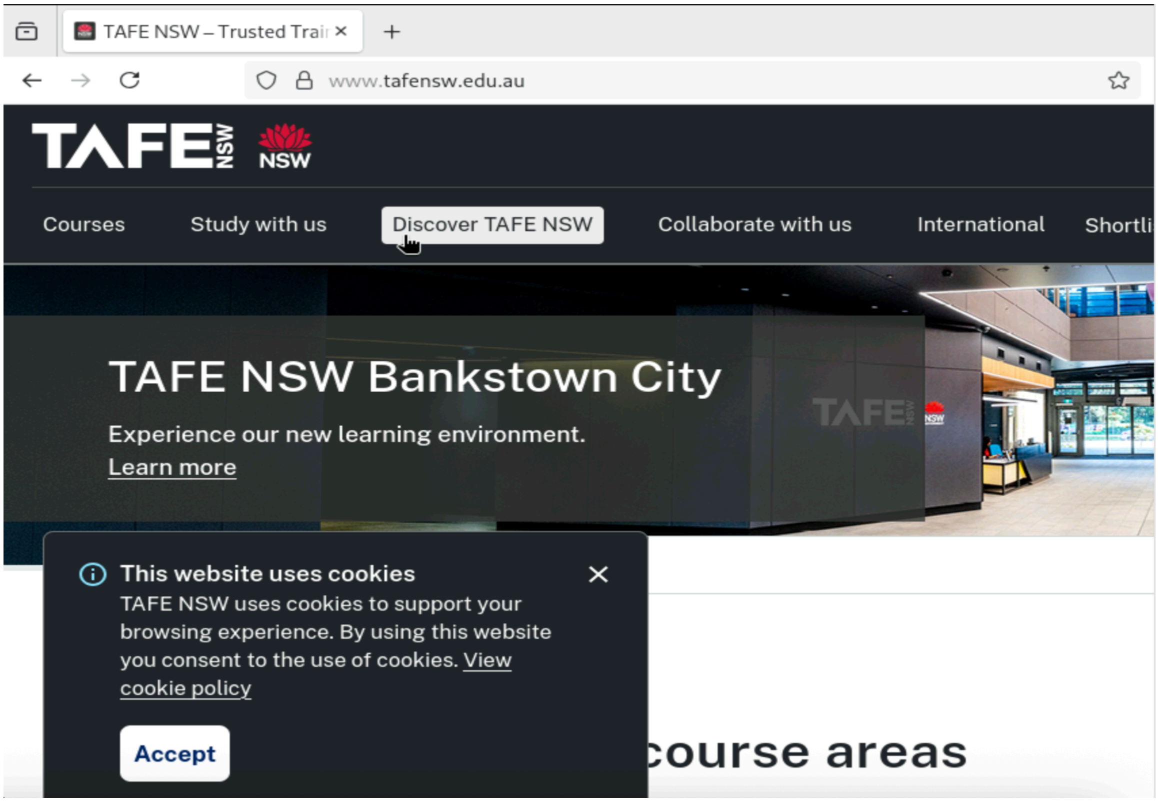
Task 2: Connecting to the Nested VM

- Started the VM.
- Waited for full boot.
- Logged in using student/student credentials.



Task 3: Testing Internet Connectivity

- Opened Firefox.
- Navigated to www.tafensw.edu.au.
- Confirmed successful loading.



Conclusion

In this lab, I successfully installed and configured a nested Debian Linux VM in Hyper-V, verified network connectivity, and applied essential VM settings. This lab reinforced my understanding of Linux-based virtual environments for cybersecurity purposes. I confirm that all work is my own.